

Arun Reddy Yeduguru

Data Engineer

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SUMMARY

- **Data Engineer** with around **5** years of experience in building scalable **data pipelines**, **ETL workflows**, and real-time streaming solutions for **financial** and **risk management** domains using **Python**, **SQL**, **PySpark**, and **Apache Kafka**.
- Expertise in cloud-based data engineering with **AWS (Glue, Redshift, S3, Lambda, Kinesis)** and **Azure (Data Factory, Synapse, Databricks)** to enable efficient data ingestion, transformation, and analytics.
- Implemented **CI/CD pipelines** using **Jenkins**, **GitHub Actions**, and **Terraform**, automating **ETL deployments** and optimizing data warehouse performance in **Redshift**, **Snowflake**, and **Athena**.
- Strengthened **data governance** and **security** by enforcing **RBAC policies**, data lineage tracking, encryption techniques, and automated data quality checks for regulatory compliance.
- Developed business intelligence solutions using **Power BI** and **Redshift**, collaborating with cross-functional teams to deliver data-driven insights and reporting solutions.

EXPERIENCE

Data Engineer

KPMG

Nov 2023 – Present | Michigan, USA

- Built scalable data pipelines with **Python**, **Apache Airflow**, and **AWS Glue**, automating financial data ingestion and transformation, reducing manual processing time by 40%.
- Developed and optimized **ETL workflows** to support **regulatory reporting**, ensuring compliance with **SOX** and **GDPR** standards.
- Consolidated **structured and unstructured datasets** into a **centralized AWS S3 data lake**, enhancing data accessibility and reducing report generation time by 50%.
- Automated data quality checks using **PySpark** and Great Expectations, increasing data accuracy by 95% and reducing manual validation efforts by 60%.
- Designed and deployed fully automated **CI/CD** pipelines with **Jenkins**, **GitHub Actions**, **Terraform** and **AWS CloudFormation**, ensuring seamless deployment, **version control**, and infrastructure provisioning with near-zero downtime.
- Collaborated with business teams to develop **real-time dashboards** using **Power BI** and **Redshift**, accelerating decision-making by 30%.
- Optimized **query performance** on **Redshift** and **Athena**, cutting data retrieval time by 40%.
- Implemented real-time data streaming pipelines using **Apache Kafka** and **AWS Kinesis**, enabling near-instant transaction processing.

Data Engineer Associate,

Deloitte

May 2019 – Jul 2022 | India

- Designed end-to-end data pipelines for **large-scale data migration projects** using **Azure Data Factory** and **SQL Server**, improving data consistency and reliability.
- Developed **ETL** processes to support **tax advisory** and **risk management services**, ensuring high data quality for client reports.
- Integrated **Infrastructure-as-Code (IaC)** using **Terraform** and deployed **CI/CD pipelines** with **Azure DevOps & Kubernetes**, reducing deployment time by 40%.
- Optimized **ETL workflows** with **PySpark** on **Azure Databricks**, improving financial data processing speed by 45%.
- Implemented **data governance frameworks** using **Apache Atlas**, **Informatica**, and **Collibra**, ensuring 100% traceability of financial transactions.
- Built real-time data streaming workflows using **Apache Kafka**, enabling instant transaction processing and dashboard updates.
- Migrated legacy databases to **Snowflake** and **Azure Synapse**, improving query speed by 30% and reducing infrastructure costs.
- Designed and implemented **role-based access control (RBAC)** and encryption strategies for sensitive financial data on **Azure** and **Snowflake**, ensuring adherence to **GDPR** and other compliance frameworks.
- Automated data quality checks using **Python scripts** and **PySpark**, decreasing manual validation efforts by 60%.
- Implemented data lineage tracking using **Apache Atlas** and **Informatica**, ensuring 100% traceability of financial transactions.

SKILLS

Programming Languages: Python, R, SQL, Scala, Bash, Shell Scripting
ETL & Orchestration: AWS Glue, Apache Airflow, Azure Data Factory, Informatica
Big Data & Streaming: Apache Spark, Hadoop, Hive, HDFS, Apache Kafka, AWS Kinesis
Databases & Warehousing: Amazon Redshift, Snowflake, MySQL, PostgreSQL, NoSQL (MongoDB), Azure Synapse
Cloud Technologies: AWS (EC2, S3, RDS, Lambda, Glue, Step Functions), Azure (Data Factory, Synapse, Databricks, Data Lake)
Infrastructure as Code (IaC): Terraform, AWS CloudFormation
Data Governance & Security: RBAC, Apache Atlas, Data Lineage, Encryption, Data Quality Frameworks
CI/CD & DevOps: Jenkins, GitHub Actions, Docker, Kubernetes
BI & Reporting: Tableau, Power BI, Looker
Soft Skills: Problem-Solving, Collaboration, Stakeholder Management, Code Review, Continuous Learning

EDUCATION

University of Michigan-Dearborn, Dearborn, MI Master of Science in Data Science (CGPA: 3.96 / 4)	Aug 2022 – Apr 2024
SASTRA Deemed University, Thanjavur, India Bachelor of Technology in Computer Science and Engineering (CGPA: 8.52 / 10)	Jul 2017 – Jun 2021

CERTIFICATES

- Databricks Certified Data Engineer Professional (August 2024)
- Microsoft Certified Azure Data Engineer Associate (July 2024) IBM
- Data Science Professional Certificate (June 2020)